

FRC 4451 Automation Badge 2 Pretest

Name _____ Date _____

Answer without notes or internet. Each question has two parts.

The first sitting has no passing mark; the final sitting requires a 90%. Show your reasoning.

If you do not understand the question or want to pass, you may write “not sure.”

1. Ownership

A configuration object holds a mutable list.

(a) Explain why returning that list from a getter may let another class change the configuration.

(b) Describe one way to prevent that unwanted change.

Part (a):

Part (b):

2. IO and Generated Code

A project has real, simulation, and replay implementations of one IO interface when using AdvantageKit as the logging framework.

(a) Explain what changes between those modes and what the component should keep doing the same way.

(b) Explain how a call to a Lombok-generated record builder can be compiled even though its method body is absent from the handwritten source.

Part (a):

Part (b):

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3. Tests and Validation

A configuration accepts a finite tolerance of zero or greater. It rejects negative values, NaN, and either infinity.

(a) Give one accepted input and one rejected input for a test.

(b) Explain how the test proves that the invalid input was rejected.

Part (a):

Part (b):

4. Units and Frames

An elevator height is a Distance; a camera mount is represented by a robotToCamera Transform3d.

(a) Explain the different information those values represent.

(b) Explain why a robot-relative displacement cannot be added directly to a field-relative position without checking the coordinate frame.

Part (a):

Part (b):

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5. Control and Conversion

A motor turns through 12 rotor rotations. The mechanism has a 3:1 reduction, meaning three rotor turns produce one mechanism turn.

- (a) Calculate the mechanism turns and explain where the ratio is applied.
- (b) Distinguish an open-loop voltage request from a closed-loop velocity goal.

Part (a):

Part (b):

6. Interlocks and Commands

A parent subsystem owns an elevator, pivot, and conveyor. The conveyor may run only when the elevator and pivot are both at an approved measured position.

- (a) Describe what a refused conveyor request should do and what evidence should be recorded.
- (b) Explain why a command using these components requires the parent subsystem.

Part (a):

Part (b):

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7. Path Following

A path follower receives a pose supplier and a drive-output consumer.

(a) Explain the direction of information through each callback and why the pose should be read when needed.

(b) Explain what can go wrong if a mechanism event command also requires the drivetrain that the autonomous command is using.

Part (a):

Part (b):

8. Vision Observations

Two cameras produce timestamped observations. One camera sometimes produces no usable pose.

(a) Describe how code should handle the missing result and ensure each accepted observation is applied once, with timestamp ordering tracked per camera.

(b) Explain what robotToCamera describes and why each camera needs its own transform.

Part (a):

Part (b):

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9. Vision Filtering

A single-tag vision pose and the current estimated pose use the same field frame and metres.

- (a) Describe how to compute the translation residual and its magnitude.
- (b) Explain the difference between rejecting an observation with a named predicate and choosing uncertainty for an accepted observation.

Part (a):

Part (b):

10. Shared State and Replay

A robot-state object stores pose and derives a readiness fact.

- (a) Explain why it may expose that fact to a trigger without directly controlling a motor.
- (b) Describe the evidence that a saved log successfully replayed through the current code, including how a new diagnostic output helps verify the result.

Part (a):

Part (b):
